

CS12320 Main Individual Assignment: Patience Card Game

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Introduction

This project documents my Java implementation of a Patience card game, a solitaire-style game where players strategically combine cards to create a single pile of matching cards and suits.

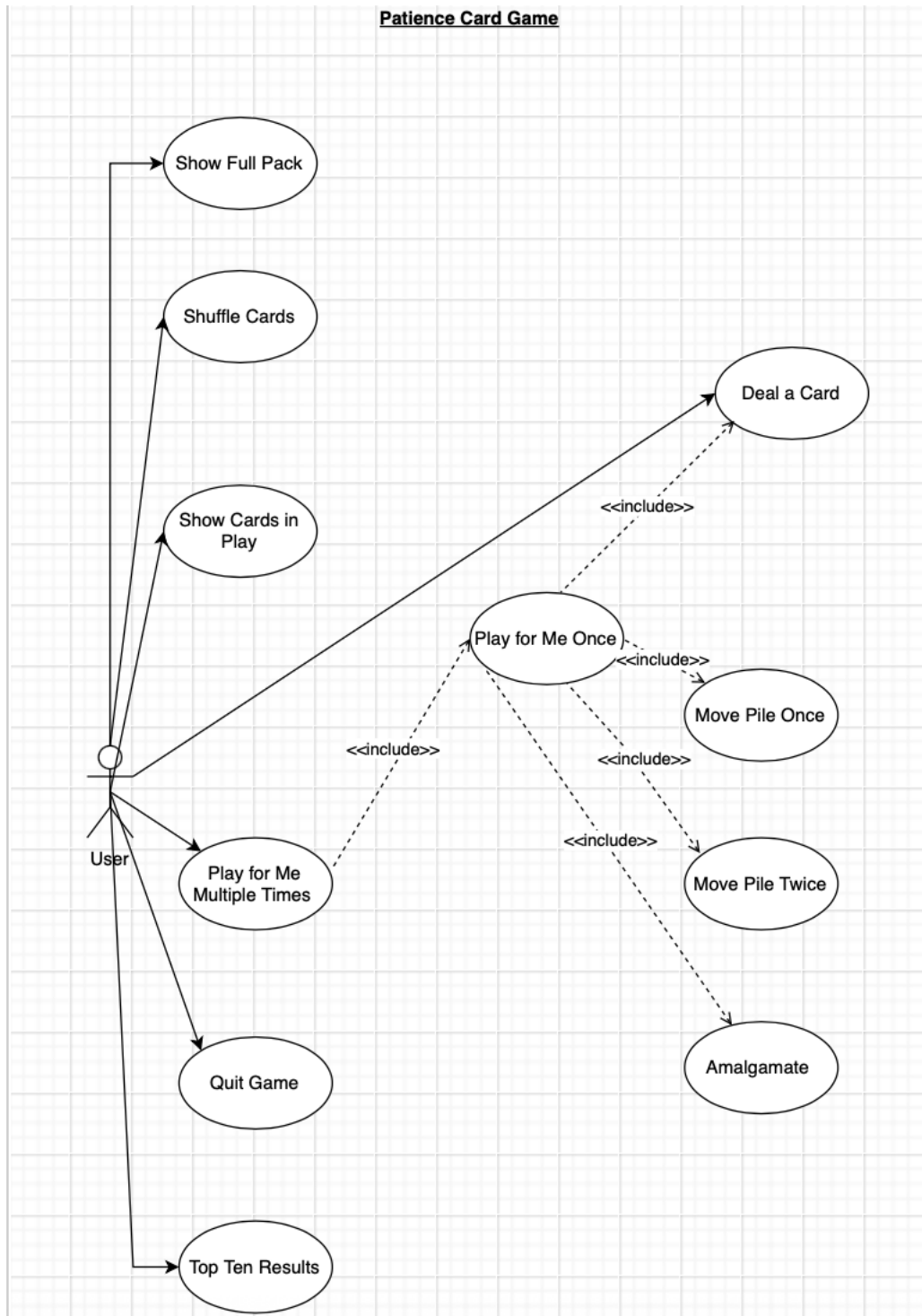
The application features both a command-line interface for gameplay actions (viewing deck, shuffling, dealing, making moves) and a graphical interface displaying the game visually. Player scores are tracked in a text-based database.

A package structure separates model components from JavaFX visual elements. The Game class integrates CLI and GUI functionality, while the Application class provides CLI-only operation.

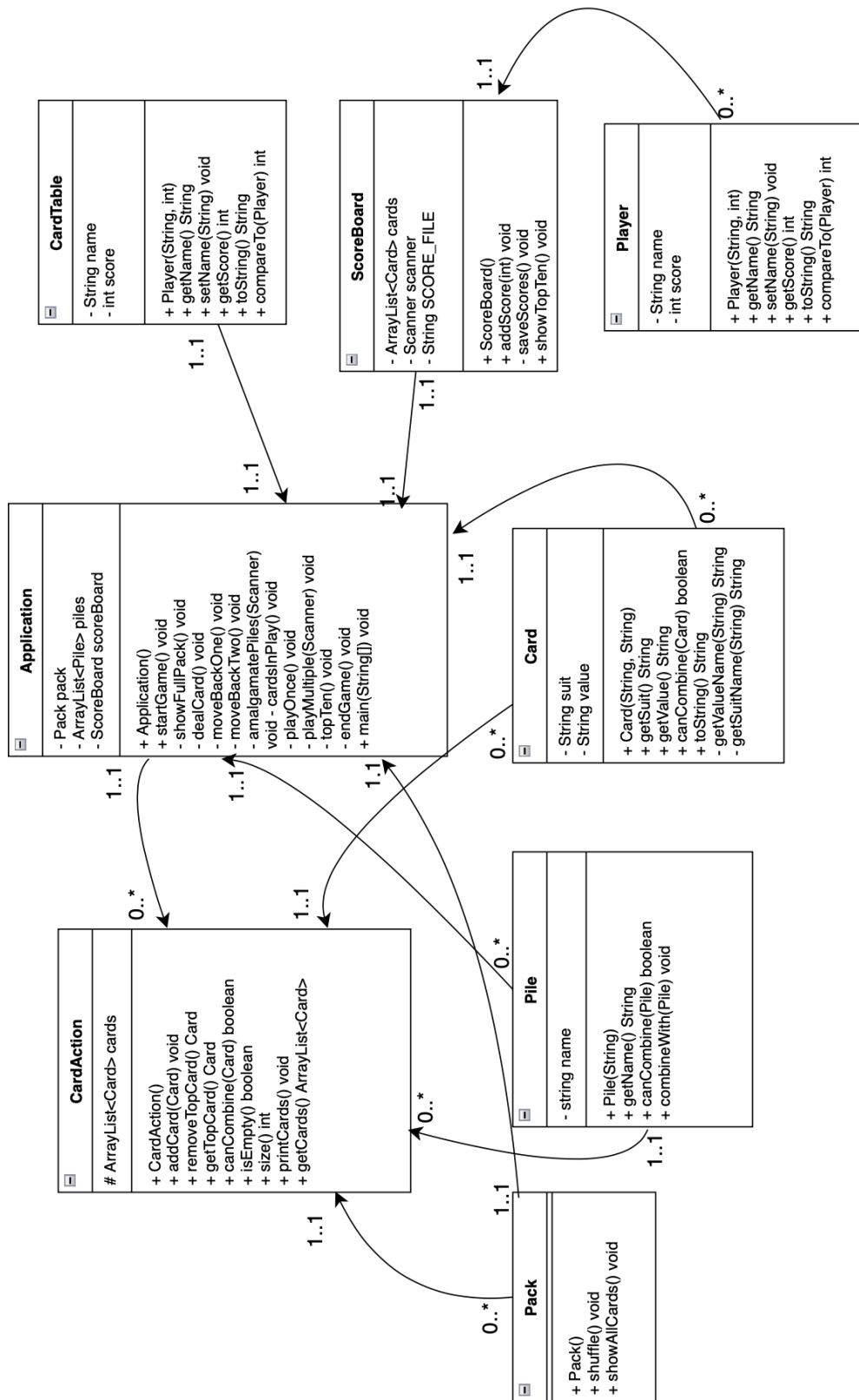
This report includes design documentation with UML diagrams, implementation details, testing methodology, and evaluation. All functional (FR1-FR11) and non-functional requirements (NFR1-NFR3) have been successfully implemented and tested as specified.

Design

Use Case Diagram



UML Diagram



Class Description

Application

This class is the main CLI interface for the card game. It manages the game state and user interface through a menu interface. T

- **main** - Program start point.
- **startGame** - Main control loop displaying menu and processing user input.
- **showFullPack** - Displays all cards in the deck.
- **dealCard** - Draws a card from pack and creates a new pile.
- **moveBackOne** - Moves last pile onto previous one if possible.
- **moveBackTwo** - Moves last pile over two positions if possible.
- **amalgamatePiles** - Combines two user-selected piles.
- **cardsInPlay** - Shows all current piles and their top cards.
- **playOnce** - Makes best possible move automatically.
- **playMultiple** - Makes multiple automatic moves sequentially.
- **topTen** - Displays best scores achieved so far.
- **endGame** - Finalises game and records score.

CardAction

This abstract class provides common functions for collections of cards. It is a parent class for Pack and Pile classes.

- **CardAction()** - Constructor that initialises an empty ArrayList to store cards.
- **addCard(Card card)** - Adds a card to the collection at the end of the ArrayList.
- **removeTopCard()** - Removes and returns the top card from the collection. Returns null if empty.
- **getTopCard()** - Returns the top card without removing it. Returns null if empty.
- **canCombine(Card card)** - Checks if the top card of this collection can be combined with the specified card.
- **isEmpty()** - Returns true if the collection contains no cards.
- **size()** - Returns the number of cards in the collection.
- **printCards()** - Displays all cards in the collection on the command line.
- **getCards()** - Returns a new ArrayList containing copies of all cards in the collection.

Pack

This class extends CardAction and represents a complete deck of 52 playing cards.

- **Pack()** - Constructor initialises a standard 52-card deck with all combinations of suits and values.
- **shuffle()** - Randomises the order of cards using Java's Collections.shuffle.

- **showAllCards()** - Displays all cards in the pack with numbering and shows the total count.

Pile

This class extends CardAction and represents a pile of cards in play. Each pile has a name and can be combined with other piles based if compliant with rules.

- **Pile(String name)** - Constructor that creates an empty pile with the specified name.
- **getName()** - Returns the name of the pile.
- **canCombine(Pile other)** - Checks if this pile can be combined with another pile based on their top cards.
- **combineWith(Pile pile)** - Merges another pile into this one by adding all its cards and clearing the source pile.

Player

This class represents a person playing the game, with attributes for name and score. It enables player scores to be saved and loaded which also allows score comparison.

- **Player(String name, int score)** - Constructor that creates a player record with the given name and score.
- **getName()** - Returns the player's name.
- **setName(String name)** - Updates the player's name.
- **getScore()** - Returns the player's score (number of piles at game end).
- **toString()** - Returns a formatted string representation of the player with name and score.
- **compareTo** - Compares players based on score, enabling sorting for the scoreboard.

ScoreBoard

This class tracks the scores of each player and saves results to a text base data base.

- **ScoreBoard()** - Constructor that initialises an empty players list and scanner for input.
- **addScore(int pileCount)** - Records a game result by prompting for player name and creating a Player object.

- **saveScores()** - Private helper method that writes all player scores to the scoreBoard.txt file.
- **showTopTen()** - Displays the top ten (or fewer) player score, from lowest to highest

Pseudo Code – Amalgamate

Amalgamate method is the most complex in my project as it handles user input to ensure they have selected a valid option. It ensures the card rules are being met to allow card combination. If piles are removed, it must ensure remaining pile do not shift order and provides error feedback for the user as well as updating GUI if required.

```

FUNCTION amalgamatePiles(scanner)
    IF piles.size < 2 THEN
        PRINT "Need at least two piles to amalgamate!"
        RETURN
    END IF

    PRINT "Enter first pile number (1-" + piles.size + "): "
    firstPileIndex = Integer.parseInt(scanner.nextLine()) - 1

    PRINT "Enter second pile number (1-" + piles.size + "): "
    secondPileIndex = Integer.parseInt(scanner.nextLine()) - 1

    IF firstPileIndex < 0 OR firstPileIndex >= piles.size OR
       secondPileIndex < 0 OR secondPileIndex >= piles.size OR
       firstPileIndex = secondPileIndex THEN
        PRINT "Invalid pile numbers!"
        RETURN
    END IF

    firstPile = piles.get(firstPileIndex)

```



```
secondPile = piles.get(secondPileIndex)
```

```
IF firstPile.canCombine(secondPile) THEN
```

```
    // Combine piles
```

```
    firstPile.combineWith(secondPile)
```

```
    // Remove second pile
```

```
    piles.remove(secondPileIndex)
```

```
    PRINT "Successfully amalgamated piles."
```

```
ELSE
```

```
    PRINT "Cannot combine these piles - no matching suit or value!"
```

```
END IF
```

```
END FUNCTION
```

Testing

ID	Requirement	Description	Input	Output	Reference	P/F
2.1.0	FR1: Show the full pack	Display all cards in the deck	Menu Choice = "1"	All 52 cards listed with their suit and value	Screenshot 2.1.0	P
2.2.0	FR2: Shuffle the cards	Randomise the order of cards in the deck	Menu Choice = "2"	"Cards have been shuffled!" message	Screenshot 2.2.0	P
2.3.0	FR3: Deal a card	Deal the top card from the deck to create a new pile	Menu Choice = "3"	New pile created with the top card from the deck	Screenshot 2.3.0	P
2.3.1	FR3: Deal when deck is empty	Attempt to deal when no cards remain	Menu Choice = "3" after dealing all 52 cards	"No remaining cards to deal!" message	Screenshot 2.3.1	P
2.4.0	FR4: Move last pile onto previous one (valid)	Combine last pile with previous one when they match	Menu Choice = "3" (deal multiple cards until matching cards appear), then "4"	Successfully moved last pile onto previous one" message	Screenshot 2.4.0	P
2.4.1	FR4: Move last pile onto previous one (invalid)	Attempt to combine non-matching piles	Menu Choice = "3" (deal multiple cards until non-matching cards	"Can't combine these piles - no matching suit or value!" message	Screenshot 2.4.1	P

			appear), then "4			
2.4.2	FR4: Move last pile when fewer than 2 piles	Attempt to use FR4 with insufficient piles	Deal one card, then Menu Choice = "4"	"Need at least two piles to perform this move!" message	Screenshot 2.4.2	P
2.5.0	FR5: Move last pile back over two piles (valid)	Combine last pile with pile 3 positions back when they match	Menu Choice = "3" (deal multiple cards until matching cards appear), then "5"	"Successfully moved last pile back over two piles." message	Screenshot 2.5.1	P
2.5.1	FR5: Move last pile back over two piles (invalid)	Attempt to combine non-matching piles	Menu Choice = "3" (deal multiple cards until non-matching cards appear), then "5"	"Cannot combine these piles - no matching suit or value!" message	Screenshot 2.5.1	P
2.5.2	FR5: Move last pile when fewer than 3 piles	Attempt to use FR5 with insufficient piles	Deal two cards, then Menu Choice = "5"	"Need at least three piles to perform this move!" message	Screenshot 2.5.2	P
2.6.0	FR6: Amalgamate piles in the middle (valid)	Combine two specified piles when they match	Deal multiple cards, then Menu Choice = "6", Pile numbers = valid matching pile numbers	"Successfully amalgamated piles." message	Screenshot 2.6.0	P
2.6.1	FR6: Amalgamate piles in the middle (invalid)	Attempt to combine non-matching piles	Deal multiple cards, then Menu Choice = "6", Pile numbers = valid non-matching pile numbers	"Cannot combine these piles - no matching suit or value!" message	Screenshot 2.6.1	P
2.6.2	FR6: Amalgamate with invalid pile number	Attempt to combine using invalid pile numbers	Deal multiple cards, then Menu Choice = "6", Pile numbers =	"Invalid pile numbers!" message	Screenshot 2.6.2	P

			invalid numbers (e.g., 0, negative, or beyond pile count)			
2.7.0	FR7: Show all cards in play	Display the current state of all piles	Deal multiple cards, then Menu Choice = "7"	List of all piles with their top cards	Screenshot 2.7.0	P
2.7.1	FR7: Show cards when no cards in play	Attempt to show cards when no cards have been dealt	Menu Choice = "7" before dealing any cards	"No cards in play yet!" message	Screenshot 2.7.1	P
2.8.0	FR8: Play for me once (move over two)	Auto-play a single move preferring two-pile jumps	Deal cards with a valid two-pile jump possible, then Menu Choice = "8"	"Auto-played: Moved last pile back over two piles." message	Screenshot 2.8.0	P
2.8.1	FR8: Play for me once (game over)	Auto-play when no moves and no cards left	Deal all cards, make moves until no more are possible, then Menu Choice = "8"	"No valid moves and no cards left in the pack!" message	Screenshot 2.8.4	P
2.9.0	FR9: Play for me a number of times	Auto-play multiple moves in sequence	Deal cards, then Menu Choice = "9", Number of moves = 5	"No valid moves 5 sequential auto-play operations	Screenshot 2.9.0	P
2.9.1	FR9: Play until completion	Auto-play until no more moves are possible	Deal cards, then Menu Choice = "9", Number of moves = large number (e.g., 100)	Auto-plays until "No more valid moves possible" message	Screenshot 2.9.1	P
2.10.0	FR10: Show top ten results	Display the best scores saved so far	Play a game to completion, then Menu Choice = "10"	Sorted list of best scores (lowest pile counts first)	Screenshot 2.10.0	P
2.10.1	FR10: Show results when none exist	Attempt to show scores	Menu Choice = "10" before	"No scores saved yet!" message	Screenshot 2.10.1	P

		before any are saved	playing any games			
2.11.0	FR11: Quit with active game	Exit the game after playing	partial game, then Menu Choice = "Q"	Game ends, score is saved	Screenshot 2.11.0	P
2.11.1	FR11: Quit without active game	Exit the game without playing	Menu Choice = "Q" before dealing any cards	Game ends without score prompts	Screenshot 2.11.1	P

Testing Screenshots

Screenshot - 2.1.0

```
====PATIENCE CARD GAME MENU====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 1
Cards in the pack:
1. 2 of Hearts
2. 3 of Hearts
3. 4 of Hearts
4. 5 of Hearts
5. 6 of Hearts
6. 7 of Hearts
7. 8 of Hearts
8. 9 of Hearts
9. 10 of Hearts
10. Jack of Hearts
11. Queen of Hearts
12. King of Hearts
13. Ace of Hearts
14. 2 of Diamonds
15. 3 of Diamonds
16. 4 of Diamonds
17. 5 of Diamonds
18. 6 of Diamonds
19. 7 of Diamonds
20. 8 of Diamonds
21. 9 of Diamonds
22. 10 of Diamonds
23. Jack of Diamonds
24. Queen of Diamonds
25. King of Diamonds
26. Ace of Diamonds
27. 2 of Clubs
28. 3 of Clubs
29. 4 of Clubs
30. 5 of Clubs
31. 6 of Clubs
32. 7 of Clubs
33. 8 of Clubs
34. 9 of Clubs
35. 10 of Clubs
36. Jack of Clubs
37. Queen of Clubs
38. King of Clubs
39. Ace of Clubs
40. 2 of Spades
41. 3 of Spades
42. 4 of Spades
43. 5 of Spades
44. 6 of Spades
45. 7 of Spades
46. 8 of Spades
47. 9 of Spades
48. 10 of Spades
49. Jack of Spades
50. Queen of Spades
51. King of Spades
52. Ace of Spades
Total cards in the pack: 52
```

Screenshot - 2.2.0

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 2
Cards have been shuffled!
```

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: |
```

Screenshot - 2.3.0

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 3
Dealt: 7 of Spades
Created new pile: Pile 1

=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: |
```

Screenshot - 2.3.1

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 3
Dealt: Queen of Diamonds
Created new pile: Pile 52

=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 3
No remaining cards to deal!
```


Screenshot - 2.4.0

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 3
Dealt: 8 of Spades
Created new pile: Pile 5
```

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 4
Successfully moved last pile onto previous one
```

Screenshot - 2.4.1

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 4
Can't combine these piles - no matching suit or value!

=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: |
```

Screenshot - 2.4.2

```
====PATIENCE CARD GAME MENU====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 4
Need at least two piles to perform this move!

====PATIENCE CARD GAME MENU====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: |
```

Screenshot 2.5.0

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 3
Dealt: 8 of Diamonds
Created new pile: Pile 4

=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 5
Successfully moved last pile back over two piles.
```

Screenshot - 2.5.1

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 5
Cannot combine these piles - no matching suit or value!

=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: |
```

Screenshot - 2.5.2

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 5
Need at least three piles to perform this move!

=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: |
```

Screenshot - 2.6.0

```
====PATIENCE CARD GAME MENU====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 7
Cards currently in play:
  Pile 1 top card: Ace of Spades
  Pile 2 top card: King of Spades
  Pile 3 top card: 8 of Hearts
  Pile 4 top card: 3 of Clubs
  Pile 5 top card: 2 of Diamonds
  Pile 6 top card: Queen of Diamonds

====PATIENCE CARD GAME MENU====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 6
Enter first pile number (1-6): 1
Enter second pile number (1-6): 2
Successfully amalgamated piles.
```

Screenshot - 2.6.1

=====PATIENCE CARD GAME MENU=====

1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
- Q. Quit

Enter option: 6

Enter first pile number (1-5): 3

Enter second pile number (1-5): 4

Cannot combine these piles - no matching suit or value!

=====PATIENCE CARD GAME MENU=====

1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
- Q. Quit

Enter option: |

Screenshot - 2.6.2

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 6
Enter first pile number (1-5): 0
Enter second pile number (1-5): 5
Invalid pile numbers!
```

Screenshot - 2.7.0

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 7
Cards currently in play:
  Pile 1 top card: 7 of Clubs
  Pile 2 top card: 10 of Clubs
  Pile 3 top card: 10 of Spades
  Pile 4 top card: 6 of Spades

=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option:
```

Screenshot - 2.7.1

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 7
No cards in play yet!
```

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option:
```

Screenshot - 2.8.0

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 8
Auto-played: Moved last pile back over two piles.

=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option:
```

Screenshot - 2.8.1

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 8
Auto-played: Amalgamated pile 2 with pile 3

=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 8
No valid moves and no cards left in the pack!
Game over! Final score: 3 piles
=====GAME OVER=====
Your score (piles remaining): 3
Score Saved
Enter your name: 8
Scores saved to scoreBoard.txt
Score recorded for 8: 3 piles
```

Screenshot - 2.9.0

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 9
How many moves would you like to make? 5
Move 1:
Auto-played: Amalgamated pile 1 with pile 2
Move 2:
Dealt: 6 of Hearts
Created new pile: Pile 3
Auto-played: No valid moves, dealt a new card.
Move 3:
Auto-played: Moved last pile onto previous one.
Move 4:
Dealt: 9 of Hearts
Created new pile: Pile 3
Auto-played: No valid moves, dealt a new card.
Move 5:
Auto-played: Moved last pile onto previous one.
```

Screenshot - 2.9.1

```
=====PATIENCE CARD GAME MENU=====
```

1. Show the full pack
 2. Shuffle the pack
 3. Deal a card
 4. Make a move, move last pile onto previous one
 5. Make a move, move last pile back over two piles
 6. Amalgamate piles in the middle (by giving their numbers)
 7. Show all cards in play
 8. Play for me once
 9. Play for me a number of times
 10. Show top ten results
- Q. Quit

Enter option: 9

How many moves would you like to make? 50

Move 1:

Dealt: 5 of Hearts

Created new pile: Pile 5

Auto-played: No valid moves, dealt a new card.

Move 2:

Auto-played: Moved last pile onto previous one.

Move 3:

Dealt: Queen of Hearts

Created new pile: Pile 5

Auto-played: No valid moves, dealt a new card.

Move 4:

Auto-played: Moved last pile onto previous one.

Move 5:

Dealt: Ace of Diamonds

Created new pile: Pile 5

Auto-played: No valid moves, dealt a new card.

Move 6:

Auto-played: Amalgamated pile 1 with pile 5

Move 7:

Dealt: 2 of Hearts

Created new pile: Pile 5

Move 7:

Dealt: 2 of Hearts

Created new pile: Pile 5

Auto-played: No valid moves, dealt a new card.

Move 8:

Auto-played: Moved last pile back over two piles.

Move 9:

Auto-played: Moved last pile onto previous one.

Move 10:

Dealt: 9 of Clubs

Created new pile: Pile 4

Auto-played: No valid moves, dealt a new card.

Move 11:

Dealt: 3 of Hearts

Created new pile: Pile 5

Auto-played: No valid moves, dealt a new card.

Move 12:

Auto-played: Moved last pile back over two piles.

No more valid moves possible.

```
=====PATIENCE CARD GAME MENU=====
```

1. Show the full pack
 2. Shuffle the pack
 3. Deal a card
 4. Make a move, move last pile onto previous one
 5. Make a move, move last pile back over two piles
 6. Amalgamate piles in the middle (by giving their numbers)
 7. Show all cards in play
 8. Play for me once
 9. Play for me a number of times
 10. Show top ten results
- Q. Quit

Enter option: |

Screenshot - 2.10.0

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times|
10. Show top ten results
Q. Quit
Enter option: 10

==== TOP 10 SCORES ====
(Lower scores are better)
1. Jack Score: 1
2. Sophia Score: 1
3. Olivia Score: 2
4. Noah Score: 2
5. James Score: 3
6. Emma Score: 3
7. Ethan Score: 4
8. Jackson Score: 5
9. Ava Score: 6
10. Lucas Score: 7
```


Screenshot - 2.10.1

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: 10
No scores saved yet!
```

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: |
```

Screenshot - 2.11.0

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: q
You Win! Game completed with just 1 pile!
=====GAME OVER=====
Your score (piles remaining): 1
SUCCESSFUL GAME! You reached the minimum possible piles!
Enter your name: Jack
Scores saved to scoreBoard.txt
Score recorded for Jack: 1 piles
Game Over!

Process finished with exit code 0
```

Screenshot - 2.11.1

```
=====PATIENCE CARD GAME MENU=====
1. Show the full pack
2. Shuffle the pack
3. Deal a card
4. Make a move, move last pile onto previous one
5. Make a move, move last pile back over two piles
6. Amalgamate piles in the middle (by giving their numbers)
7. Show all cards in play
8. Play for me once
9. Play for me a number of times
10. Show top ten results
Q. Quit
Enter option: q
Game Over!

Process finished with exit code 0
```

Evaluation

Approach to Solving the Assignment

I found it difficult to implement middle pile amalgamation as I had to manage array indices when removing piles, this was hard to code. To improve I can use the GUI to display all the cards in the pile instead of just the top pile improving user experience. Furthermore, I improved my understand of abstraction and inheritance due to CardAction superclass being used.

Implementation Difficulties

I found it difficult to implement middle pile amalgamation as I had to manage array indices when removing piles, this was hard to code.

Areas for Improvement

To improve I can use the GUI to display all the cards in the pile instead of just the top pile improving user experience.

Learning Outcomes

Furthermore, I improved my understand of abstraction and inheritance due to CardAction superclass being used.

Self-Assessment

This project approximately deserves 80% as I implement all the NFR and FR functions required as well using a difficult autoplay system that selects optimal moves by first checking for two-pile jumps, then adjacent moves, and finally middle pile combinations. The ScoreBoard class with text file persistence shows attention to NFR2, while auto GUI updates after each move fulfil NFR3 requirements. Overall, the project demonstrates solid OOP principles which was required by the brief..